

Reproduction of karpathy/llm.c: GPT-2 124M fine-tuning on tinyshakespeare in CUDA

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Independent reproduction · Benchmark or leaderboard entry · Natural language processing

Abstract

Ran the llm.c quick start from the pinned commit: downloaded the starter pack (GPT-2 124M bf16 weights, GPT-2 tokenizer, tokenized tinyshakespeare) with dev/download_starter_pack.sh, built train_gpt2cu with make (bf16, no cuDNN, single GPU, compute capability 12.0, CUDA 12.8 toolkit from conda-forge), and ran ./train_gpt2cu with its default settings: one epoch (74 steps) of fine-tuning at B=4, T=1024, AdamW lr 3e-4, with validation loss every 20 steps and sample generations. Training loss per step, validation loss, average iteration time and throughput are parsed from the program output. Single NVIDIA RTX 5060 Ti 16 GB.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	GPT-2 124M tinyshakespeare	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: GPT-2 124M tinyshakespeare

Run	Metric	Final value	Step
1	val_loss	3.50806	74
1	train_loss	3.39568	74

Run	Inline claim	Value	Phase
1	final_val_loss	3.50806	
1	avg_iteration_ms	125.668	
1	tokens_per_second	32597	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	14.9 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	ddf25bf583ca9045fc92295a1ed380db41ca8658a98947d44eb73fe58e21d7d0	241 KB
r1.kvbundle.zip	0b86756dddea1a345fcf61819d5fa17e41710704876aa7589f7462d266e23700	139 KB

5 Links and licence

Reproduction of: <https://github.com/karpathy/llm.c/tree/f1e2ace651495b74ae22d45d1723443fd00ecd3a>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: gpt-2, cuda, language-model, tinyshakespeare.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (GPT-2 124M tinyshakespeare), kv:2610.00006v1/r1. Data hash 6d545071c552a8e5b2e6933585f65c1f436d8da78794d22c84622af96b5165b0.

<https://kveritas.org/records/2610.00006v1/r1.pdf>